DS	**Session**	Linked List	**Question Information**	Level 1 Challenge 35

Question description

Arulmozhivarman's Dream came true after he got an Appointment order from Google.Simon's family was very happy of his achievement.

The company mentioned Basic Salary, DA, HRA with some other benefits.

But not highlighted the Gross salary in the order.

Arulmozhivarman's father wanted to know the Gross salary of his son.

Arulmozhivarman try to his gross salary from HR department. they informed that you have to get pass grade in first month entry test. the entry test has 5 questions. one of the question was, to delete alternate nodes of a Linked List. In this program we want to remove alternate nodes from the singly linked list which is start from the second node of the linked list and remove all the alternate nodes of it.

Can you help Arulmozhivarman?

Function Description

- First Take two pointers a and b.
- Let initially a points to head and b points to pointer of a ,that is second node .
- Than make the link of a to point to pointer of b and free b.
- Next is move to a to its pointer that is next node.
- Next is move b to next node of a.
- Continue loop until a and b becomes NULL.

Constraints

0<n<100

Input Format:
The First line of the input represents the linked list elements count

Output Format:
single line prints the elements after the deletion of alternate nodes of a Linked List

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
10
```

EXPECTED OUTPUT

```
1 3 5 7 9
```

Test Case 2

INPUT (STDIN)

```
20
```

EXPECTED OUTPUT

```
1 3 5 7 9 11 13 15 17 19
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void insert_Data(struct node
**head)
```

Test Case 2

KEYWORD

```
void delete_Alt(struct node
**head)
```

Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
3	300	68

Code

solution.c

```c
#include <stdio.h>
#include <stdlib.h>

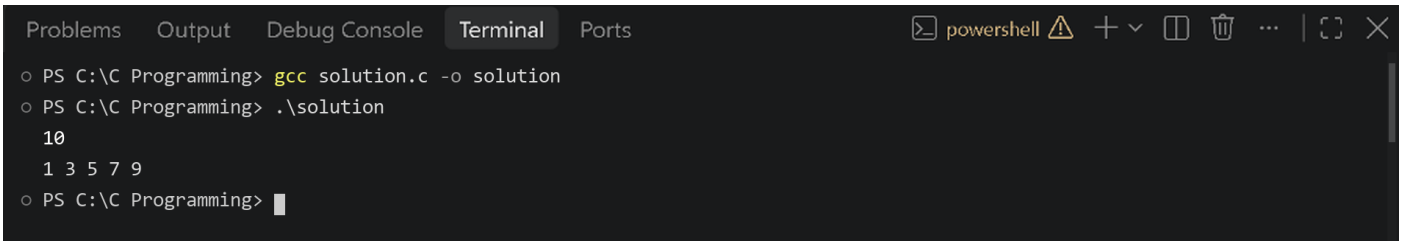
struct node {
    int data;
    struct node *next;
};

void insert_Data(struct node **head, int data) {
    struct node *newNode = (struct node*)malloc(sizeof(struct node));
    newNode->data = data;
    newNode->next = NULL;
    if (*head == NULL) *head = newNode;
    else {
        struct node *temp = *head;
        while (temp->next) temp = temp->next;
        temp->next = newNode;
    }
}

void delete_Alt(struct node **head) {
    if (*head == NULL) return;
    struct node *a = *head;
    struct node *b = (*head)->next;
    while (a && b) {
        a->next = b->next;
        free(b);
        a = a->next;
        if (a) b = a->next;
    }
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    struct node *head = NULL;
    for (int i = 1; i <= n; i++) {
        insert_Data(&head, i);
    }
    delete_Alt(&head);
    struct node *cur = head;
    while (cur) {
        printf("%d%s", cur->data, cur->next ? " " : "");
        cur = cur->next;
    }
    printf("\n");
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
10
1 3 5 7 9
PS C:\C Programming>